INCH-POUND

MIL-PRF-25732D
w/Amendment 1
23 April 2021
SUPERSEDING
MIL-PRF-25732D
29 January 2021

PERFORMANCE SPECIFICATION

PACKING, PREFORMED, PETROLEUM HYDRAULIC FLUID RESISTANT, LIMITED SERVICE AT 250 °F (121 °C)

MIL-PRF-25732D w/Amendment 1 remains inactive for new design and is no longer used, except for replacement purposes.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.

This specification covers an Acrylonitrile-Butadiene (NBR) rubber in the form of molding rings, compression seals, O-ring cord, and molded-in-place gaskets for aeronautical and aerospace applications.

Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, 2145 Monahan Way, Bldg. 28, Wright-Patterson AFB OH, 45433-7017 or e-mailed to ENGINEERING.STANDARDS@US.AF.MIL. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1.2 Application.

This material has resistance to petroleum based hydraulic fluids conforming to MIL-PRF-5606, MIL-PRF-87257 and MIL-PRF-83282, but usage is not limited to such applications. This material is suitable for use in hydraulic systems in accordance with MIL-G-5514 and SAE AS5440 and has a typical service temperature range of -65 °F to 250 °F (-54 °C to 121 °C). The service temperature range of the material is a general temperature range, but the presence of particular fluids and specific design requirements may modify this range. Each application should be considered separately. It is the responsibility of the user to determine that this specification is appropriate for the environments (temperature range, fluid exposure, etc.) in which it is sought to be used. A material providing better 275 °F aging is covered by SAE-AMS-P-83461.

1.3 Safety - hazardous materials.

While the materials, methods, applications, and processes described or referenced in this specification may involve the use of hazardous materials, this specification does not address the hazards which may be involved in such use. It is the sole responsibility of the user to ensure familiarity with the safe and proper use of any hazardous materials and to take necessary precautionary measures to ensure the health and safety of all personnel involved.

2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-G-5514	Gland Design; Packings, Hydraulic, General Requirements for
MIL-PRF-5606	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes (DoD adopted)

(Copies of this document are available online at http://www.asq.org/.)

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ASTM INTERNATIONAL

ASTM D297	Standard Test Methods for Rubber Products - Chemical Analysis
ASTM D395	Standard Test Methods for Rubber Property - Compression Set
ASTM D412	Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers-Tension
ASTM D471	Standard Test Methods for Rubber Property - Effect of Liquids
ASTM D1329	Standard Test Method for Evaluating Rubber Property-Retraction at Lower Temperatures (TR Test)
ASTM D1414	Standard Test Methods for Rubber O-Rings (DoD adopted)
ASTM D2240	Standard Test Method for Rubber Property-Durometer Hardness
ASTM D7426	Standard Test Method for Assignment of the DSC Procedure for Determining Tg of a Polymer or an Elastomeric Compound

(Copies of these documents are available online at http://www.astm.org/.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 3601-3	Fluid Power Systems - O-rings - Part 3: Quality Acceptance Criteria

(Copies of this document are available online at http://www.iso.org/.)

SAE INTERNATIONAL

SAE AIR851	O-ring Tension Testing Calculations
SAE ARP3050	Suitable Test Sizes for O-ring Specifications
SAE AMS3400	Fluid, Reference for Testing Polyalphaolefin (PAO) Resistant Material
SAE AS5440	Hydraulic Systems, Military Aircraft, Design and Installation, Requirements For
SAE AS5752	Aerospace - Visual Inspection Standard for Elastomeric Sealing Elements Other than O-rings
SAE AS6414	Manufacturing Processing Requirements for Molded Elastomer Components Used in Aerospace Applications
SAE AS28775	Packing, Preformed - MS28775 O-Ring

(Copies of these documents are available online at http://www.sae.org/.)

2.4 Order of precedence.
Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific

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exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification.

The packings furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2). Qualification testing shall be performed every three years.

3.2 Compound.

Shall be prepared from ingredients as shall be necessary to achieve the requirements detailed in this specification and shall be a compound, based on butadiene-acrylonitrile (NBR) elastomer, fully cured to produce product meeting the requirements of 3.4. Material used shall be 100 percent virgin NBR elastomer. No reprocessed or non-NBR polymer is acceptable.

3.3 Dimensions and tolerances.

The dimensions and tolerances of standard size O-rings procured to this specification shall conform to SAE AS28775. Non-standard parts shall conform to applicable design drawings.

3.4 Physical properties.

The physical properties of the rubber shall conform to the values given in Table I. Material shall be tested on the O-ring SAE AS28775-214 pertaining to this material specification. Calculations of tensile strength and elongation may be made in accordance with SAE AIR851.

TABLE I. Physical properties.

Original Properties	Requirements	Test Method
As Received		
Specific gravity	±0.02 of qualification value	ASTM D1414 and ASTM D297
Hardness Durometer Type A, points	73, ±5	ASTM D1414 and ASTM D2240
Tensile strength, psi, min.	1350	ASTM D1414
Elongation, percent, min	160 (may not be lower than 160 percent in production)	ASTM D1414
Tensile Stress, min, psi	500	ASTM D1414
Temperature retraction, TR-10, 50 percent elongation and 10 percent return, °F (°C), max	-49 °F (-45 °C)	ASTM D1414 and ASTM D1329
Glass transition (Tg) Mid-Point, °F (°C), max	-49 °F (-45 °C)	ASTM D7426

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TABLE I. Physical properties - Continued.

Original Properties	Requirements	Test Method
As Received		
Short-term compression set, percent, max	55	ASTM D1414 and ASTM D395 Method B Temp: 250 ±5 ˚F (121 ±3 °C) Time: 70 ±0.5 hours
Long-term Compression set, percent, max	75	ASTM D1414 and ASTM D395 Method B Temp: 250 ±5 ˚F (121 ±3 °C) Time: 336 ±0.5 hours
Fluid MIL-PRF-5606 Resistance ASTM D471		
Hardness Durometer “Type A”, change, point, max	+5 to -15	ASTM D1414 and ASTM D471 Temp: 250 ±5 °F (121 ±3 °C) Time: 70 ±0.5 hours MIL-PRF-5606 Fluid
Tensile Strength decrease, percent, max	50	
Elongation decrease, percent max	35	
Volume change, percent, max	0.5 to 15	
Temperature retraction, TR-10, 50 percent elongation and 10 percent return, max	-49 °F (-45 °C)	
Short-term compression set, percent of original deflection, max	55	ASTM D1414 and ASTM D395 Method B Temp: 250 ±5°F (121 ±3 °C) Time: 22 ±0.5 hours MIL-PRF-5606 Fluid
Long-Term Compression set, percent of original deflection, max	75	ASTM D1414 (ASTM D471) Temp: 250 ±5 °F (121 ±3 °C) Time: 336 ±0.5 hours MIL-PRF-5606 Fluid
Fluid AMS3400 Resistance ASTM D471		
Hardness Durometer “Type A”, change, point, max	±10	ASTM D1414 and ASTM D471 Temp: 250 ±5 °F (121 ±3 °C) Time: 70 ±0.5 hours SAE AMS3400 Fluid
Tensile Strength decrease, percent, max	40	
Elongation decrease, percent, max	45	
Volume change, percent, max	0.5 to 15	
Temperature retraction, TR-10 50 percent elongation and 10 percent return, max	-49 °F (-45 °C)	
Short-term compression set, percent of original deflection, max	55	ASTM D1414 (ASTM D471) Temp: 250 ±5 °F (121 ±3 °C) Time: 22 ±0.5 hours SAE AMS3400 Fluid
Long-Term Compression set, percent of original deflection, max	75	ASTM D1414 (ASTM D471) Temp: 250 ±5 °F (121 ±3 °C) Time: 336 ±0.5 hours SAE AMS3400 Fluid

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3.5 Dynamic cycling.
When tested in accordance with 4.5, the O-rings shall withstand a minimum of 75,000 cycles without failure. Failure is determined as that number of cycles at which a test O-ring leaks a total of 70 mL of hydraulic fluid. Each test consists of two test cells containing two test O-rings each. A minimum of three test runs shall be conducted making a total of six test cells to be tested. The number of cycles until failure shall be recorded. The highest number of cycles and lowest number of cycles shall be discarded. The remaining four values shall be averaged and that value must be 75,000 minimum for the O-rings to pass the test.

3.6 Identification.
All O-rings shall be individually packaged. Each envelope shall have the following information printed on the outside:

National Stock Number (NSN)

Military Part Number

Material Specification

Manufacturer's Name

Manufacturer's Compound Number

Manufacturer's Batch Number

Contract Number

Cure Date

NOTE: A lot of a product may be packaged in small quantities and delivered under the basic lot approval provided lot identification is maintained.

4. VERIFICATION

4.1 Classification of inspections.
The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 Qualification inspection.

4.2.1 Samples.
Test samples shall be O-rings conforming to size -214 of SAE AS28775. Hardness shall be determined on hardness discs of at least 0.25 inch thickness.

4.2.2 Tests.
Qualification tests shall consist of all the tests specified in 4.3 through 4.5.

4.3 Conformance inspection.
Conformance inspection shall include tests specified in 4.3.2 and 4.3.3. The following tests shall be conducted on each batch and lot of material. Unless otherwise specified in the contract or purchase order, the manufacturer of the product shall be responsible for performing all required

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tests as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to sample and perform any testing deemed necessary to ensure that the product conforms to this specification.

4.3.1 Sampling for inspection.
Sampling for the conformance inspection shall be in accordance with ANSI/ASQ Z1.4, except where otherwise indicated herein. Conformance tests are required for all batches of material.

4.3.2 Batch.
A batch shall be the quantity pf material compounded on a mill or mixer at one time. A batch shall always include SAE AS28775-214 O-rings and hardness discs for conformance tests. A batch shall consist of a maximum of 200 pounds. Conformance tests shall be conducted on each 200 pounds if the batch size is more than 200 pounds.

TABLE II. Batch tests.

Property	Requirement	Test Method
As Received		
Specific Gravity	±0.02 of qualification value	ASTM D1414 (ASTM D297)
Hardness Durometer “Type A”, points, max	68, ±5	ASTM D1414 (ASTM D2240)
Tensile strength, psi, min	1350	ASTM D1414
Elongation, percent, min	160 (may not be lower than 160 percent in production)	ASTM D1414
Temperature retraction, TR-10 50 percent elongation and 10 percent return, °F (°C), max	-49 °F (-45 °C)	ASTM D1414 (ASTM D1329)
Fluid MIL-PRF-5606 Resistance ASTM D471		
Hardness Durometer “Type A” change, points, max	+5 to -15	ASTM D1414 (ASTM D471) Temp: 250 ±5 °F (121 ±3 °C) Time: 70 ±0.5 hours MIL-PRF-5606 Fluid
Volume change, percent, max	0.5 to 15	
Short-term compression set, percent of original deflection, maximum	55	
Temperature retraction, TR-10, 50 percent elongation and 10 percent return, °F (°C), maximum	-49 °F (-45 °C)	

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4.3.2.1 Batch testing.
SAE AS6414 defines batch testing. Testing on the same batch and state of cure are acceptable as long as the SAE AS28775-214 O-rings are cured and tested within 90 days of the manufacture of the subject lot. Batch and the blending of compound as defined in SAE AS6414. All incoming batches and or blended batches shall be tested in accordance with Table II.

4.3.3 Acceptance lot tests.
Requirements shown in Table III are acceptance lot tests and shall be performed on each lot of the O-rings being supplied. It shall be permitted to use multiple such O-rings which shall be made from the same lot and batch to perform the applicable test. O-ring sizes that are suitable for testing (tensile and elongation) are shown in SAE ARP3050.

TABLE III. Acceptance lot tests.

Property	Requirement	Test Method
As Received		
Hardness Durometer “Type A”	68, ±5	ASTM D1414 and ASTM D2240
Tensile strength, psi, min	1350	ASTM D1414
Elongation, percent, min	160 (may not be lower than 160 percent in production)	ASTM D1414
Fluid MIL-PRF-5606 Resistance ASTM D471		
Volume change, percent, max	0.5 to 15	ASTM D1414 and ASTM D471 Temp: 250 ±5 °F (121 ±3 °C) Time: 70 ±0.5 hours MIL-PRF-5606 Fluid
Short-term compression set, percent of original deflection, maximum	55	ASTM D1414 and ASTM D471 Temp: 250 ±5°F (121 ±3 °C) Time: 22 ±0.5 hours MIL-PRF-5606 Fluid

4.3.4 Test samples.
The method shall be as specified in the parts standard, drawing or purchase document. If not specified, product shall be taken at random from each lot to perform all the required testing in Tables II and III, where applicable per the part type. The number of test specimens for each requirement shall be specified in the applicable test procedure.

4.4. Rejection criteria.
A lot shall be rejected upon the failure of any sample to meet the test requirements specified herein. A lot that has been rejected may be reworked to correct the deficiencies and resubmitted for acceptance.

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4.5 Dynamic cycling.
The dynamic cycling shall be performed on a rod of seal test apparatus. The apparatus, as shown on Figure 1, is capable of evaluating O-rings under a high pressure, high temperature cycling environment until failure. The apparatus is capable of testing four O-rings simultaneously. Two test cells are run concurrently with two O-rings in each cell. A test cell is shown on Figure 2. The testing environments are as follows:

a. Test temperature - 250 °F (121 °C).
b. Pressure - 1,500 pounds per square inch gauge (psig) (10,300 kPa).
c. Stroke length - 4 inches (10.2 cm).
d. Cycling rate - 30 cycles/minute, where one cycle is defined as the combined forward and return strokes.
e. Pressure is to be applied for one complete cycle and pressure is to be reduced to zero during the next cycle
f. Hydraulic fluid - MIL-PRF-5606.
g. Tests are to be conducted without back-up rings. A steel spacer is to be used to maintain the proper gap dimension.
h. Diametric clearance between rod and cell to be maintained at 0.004 inch (0.0102 cm).
i. Surface finish on test rods are to be between 10 to 20 micro-inches.

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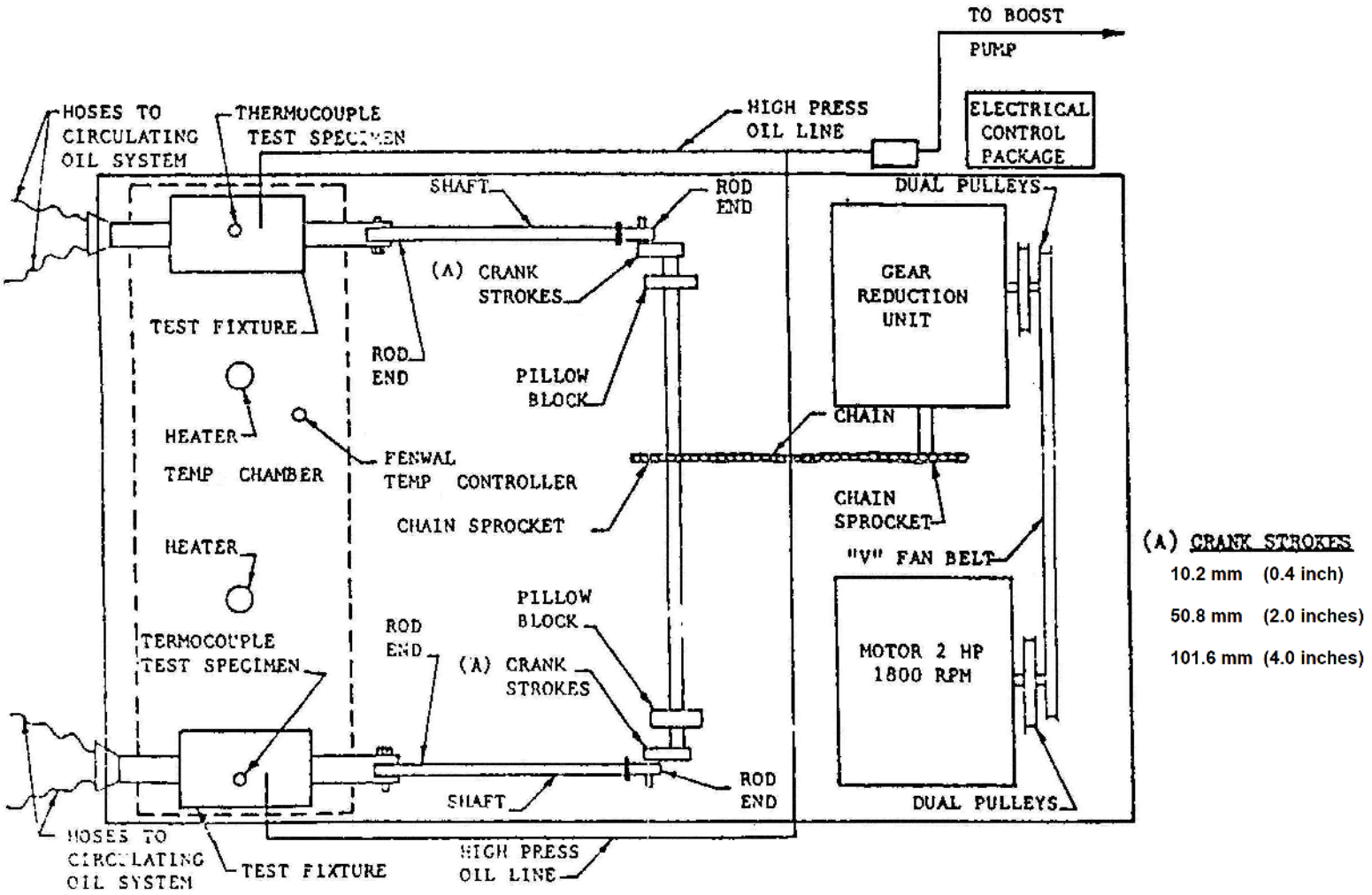


FIGURE 1. Dynamic cycling test set up.

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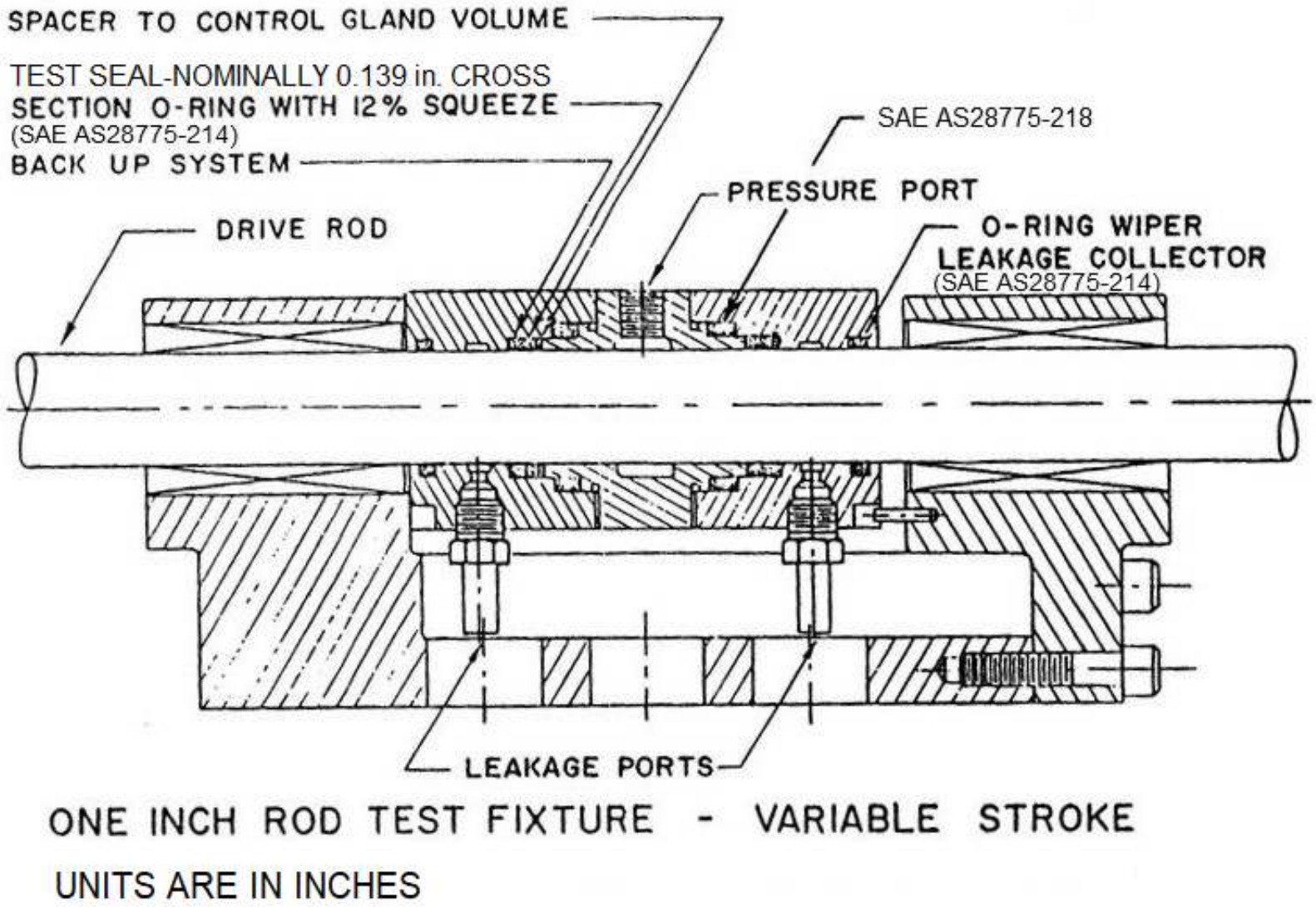


FIGURE 2. Test cell assembly.

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4.5.1 Procedure.
The procedure to be used in conducting dynamic O-ring evaluations is as follows:

a. Select four, size -214 unaged or unconditioned O-rings.

b. Lubricate the O-rings with fresh MIL-PRF-5606 hydraulic fluid and assemble the rings and spacers into the test cylinders as shown on Figure 3.

c. Rod and gland dimensions are to conform to MIL-G-5514.

d. Fill with MIL-PRF-5606 hydraulic fluid taking care to exclude entrapped air.

e. Bleed system at test cell locations to remove entrapped air in lines.

f. Check of final test cell installations in test fixtures and insure that the test cells do not translate along the rod by more than 0.005 inch (0.0127 cm).

g. Set dynamic conditions as follows:

(1) Stroke - 4 inches (101.6 mm).

(2) Rate - 30 cycles/minute.

(3) Pressure - 0 - 1500 psig (10,300 kPas).

(4) Temperature - 250 °F (121 °C).

(5) Start tests by turning on heating elements.

(6) Start test cell temperature recorder to record the temperature of each cell throughout the test.

(7) After temperature reaches the predetermined set point, start cycling and adjust temperature as required.

(8) Start leakage rate recorder at the same time that cycling is initiated.

h. Cycle O-rings at given set of conditions until failure of the rings.

i. Failure is specified as the number of cycles at which either test ring in either test cell leaks a total of 70 mL of hydraulic fluid. At this point, the test cell having the failed ring will be shut down and the remaining cell operated until one or both of the two remaining test rings fail by leaking 70 mL of hydraulic fluid.

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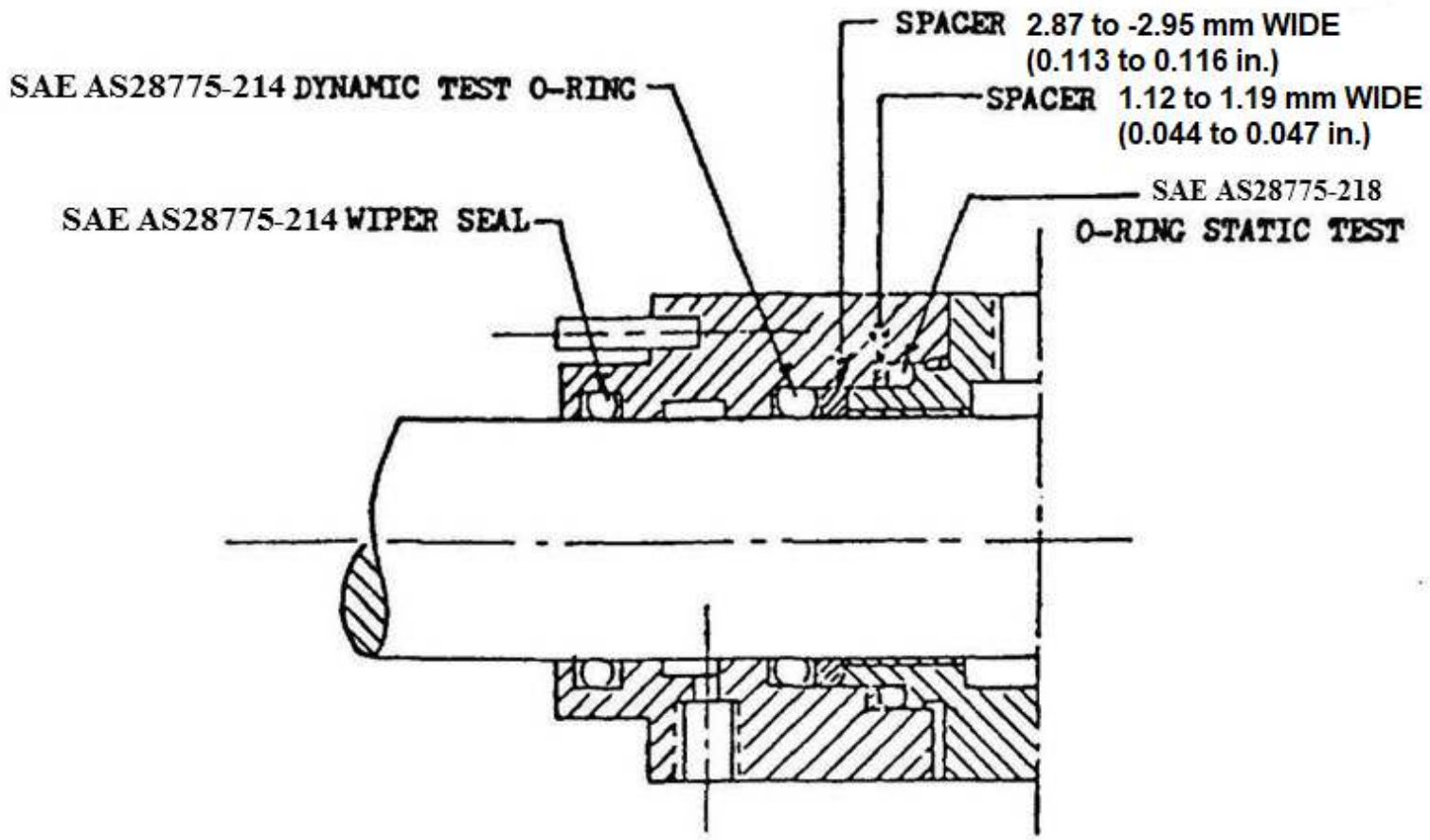


FIGURE 3. Cylinder assembly without backup rings.

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5. PACKAGING

5.1 Packaging.

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Acquisition requirements.

Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Sizes per SAE AS28775 (see 3.3).

c. Applicable National Stock Number (see 3.6).

d. Quantity (see 4.3.2)

e. Packaging requirements (see 5.1).

6.2 Qualification.

With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 25732, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from AFRL/RXSS 2179 12th Street, Wright-Patterson AFB, OH 45433-7718 or by email at Engineering.Standards@us.af.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3 Preferred testing fluid.

It is recommended that Greening ARM-201 brand reference fluid be used for the physical properties testing for aging in accordance with MIL-PRF-5606.

6.4 Subject term (key word) listing.

Butadiene Acrylonitrile
Compression
Molded Rings
NBR
Resistant
Rubber

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6.5 Amendment notations.
The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 20

Reviewer:
DLA - GS

Preparing activity:
Air Force - 20

Agent:
Air Force - 110

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